Introduction to Object Detection and Segmentation

Models, datasets, and demos, with an experimentation-first approach

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Introduction

In this handout we take an experimentation-first approach to two core problems in computer vision: object detection and segmentation. As the first handout of the course, it gives an overview of the field. Because machine learning research places a strong emphasis on reproducibility and replicability, published models are easy to test, and newcomers can quickly start contributing to the field.

Goals

- Learn about the current state of the art in object detection and segmentation from a high-level view.
- Learn about popular datasets used for these approaches.
- Run demos online to see these algorithms in action.
- Through experimentation and reading, identify some common issues with current models and existing datasets.

Definitions

Object detection

Object detection models localize each instance of a chosen set of object classes in an image. In practice, for each object we find the rectangle that encloses it (its x-coordinate, y-coordinate, width, and height) and the object's class (person, car, cat, dog, and so on). Each object is therefore described by five values.

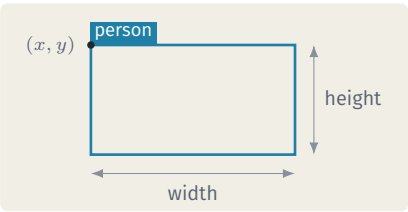


Figure 1 One detection: a box (x, y, width, height) and a class.

- **Zero-shot object detection** is an emerging approach: the model takes a list of object classes as input and detects only those, building on recent advances in large language models such as ChatGPT.

- **Pose estimation** is an important subproblem that supports many use cases today. For each object, we estimate the positions of landmark points, usually important human joints and facial landmarks, to describe the object's movements and gestures concisely.

Segmentation

Segmentation models divide a scene into regions on a per-pixel basis: for each pixel of the image, we want to find a single value.

- In **object segmentation**, we focus only on foreground objects, such as people or animals, as opposed to sidewalks and building facades.
- In **instance segmentation**, we also give each instance of an object its own identifier, so that we know which entity each pixel belongs to, in addition to its class.
- In **panoptic segmentation**, we additionally assign background pixels to their classes. This has its own uses: in a streetscape, knowing the locations of roads, sidewalks, and building facades can be crucial for mapping and spatial understanding.

Getting started with datasets

To train computer vision models, we first have to understand the datasets that exist in this space. What our models can learn is limited by what people have annotated in the past.

VisDrone DETECTION The VisDrone dataset contains aerial views of pedestrians and vehicles, collected in 14 cities in China. Samples: www.kaggle.com/datasets/wwj0510/visdrone



Figure 2 The VisDrone dataset contains aerial views of pedestrians and vehicles.

COCO SEGMENTATION AND DETECTION The COCO dataset contains a large variety of scenes and object classes, which has made it a baseline dataset for object detection and segmentation. Pretraining a model on MS-COCO gives it good priors for other real-world tasks. Samples: cocodataset.org/#explore



Figure 3 Example images from the MS-COCO dataset, with their segmentation masks.

Open Images SEGMENTATION AND DETECTION
Open Images is a dataset of 9 million images annotated with image-level labels, object bounding boxes, object segmentation masks, visual relationships, and localized narratives. Samples: storage.googleapis.com/openimages/web/visualizer/index.html

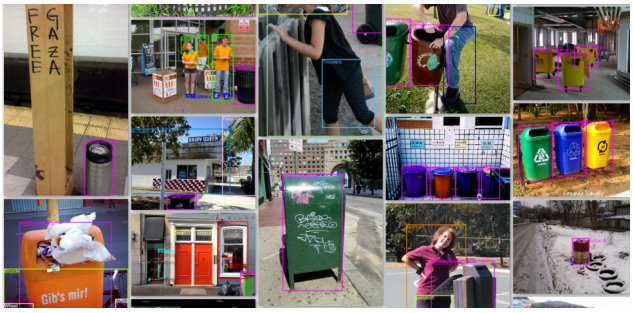


Figure 4 Examples from the Open Images dataset.

Demos

Here you will get to play with, and learn from, some state-of-the-art computer vision models, all hosted on Hugging Face Spaces.

OneFormer	A high-quality segmentation model. huggingface.co/spaces/shi-labs/OneFormer
YOLO-World	A real-time, open-vocabulary object detection model. huggingface.co/spaces/stevengrove/YOLO-World
YOLOv8 for object detection and segmentation	Performs object detection and segmentation in real time, trained on large datasets. huggingface.co/spaces/fcakyon/yolov8-segmentation
Grounding DINO and OWLv2	Slower, higher-quality open-vocabulary object detection models. huggingface.co/spaces/merve/GroundingDINO_OWL
Florence-2	A vision foundation model trained to address a range of vision tasks. huggingface.co/spaces/SkalskiP/better-florence-2

Exercises

Q1 Look online to find the classes in the COCO dataset. Can you point to any obvious classes from your daily life that are missing? What is the most similar object in the dataset?

Q2 Compare the streetscapes in the VisDrone dataset against the streetscapes of New York. What is missing? What would surprise a hypothetical entity that learned about the world only from VisDrone images?

Q3 How can we collect data from New York to build a model that will work in New Orleans, or in Boston? What are some differences in the objects between these cities?